Health Analytics Dashboard Using MongoDB

A NoSQL Approach to Multi-Source Wearable Data Integration

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ABSTRACT

Wearable fitness technologies such as WHOOP and Fitbit continuously generate large volumes of semi-structured physiological, activity, and sleep data. Integrating heterogeneous datasets from multiple devices into a single analytical platform is challenging using traditional relational databases due to rigid schema requirements and evolving data structures. This paper presents a Health Analytics Dashboard that unifies Fitbit public data (35 users, 2 months) and personal WHOOP data (2.5 years) in a MongoDB Atlas cluster, connected to an interactive Streamlit visualization layer. A Python-based ETL pipeline cleans, standardizes, and ingests all data into five MongoDB collections. The dashboard delivers activity analysis, sleep performance analytics, physiological insights, and cross-platform comparisons through six interactive tabs. Evaluation demonstrates that MongoDB's flexible document model, cloud scalability, and schema-less design are well-suited to multi-source IoT health data and outperform relational alternatives in this use case.

I. INTRODUCTION

Wearable fitness technologies, such as Whoop and Fitbit, are IoT devices that continuously collect physiological, workout, and health-related data. These devices generate large volumes of data that are stored, helping users better understand and learn more about their bodies. The Fitbit public dataset contains daily data from multiple users (35 users) over a

2-month period, while WHOOP provides my own personal data from the past 2.5 years. As a result, the data is naturally diverse, mixed, semi-structured, and irregular.

Integrating multiple datasets from heterogeneous wearable sources into a single analytical system presents a significant technical challenge: the data do not conform to a uniform structure. Traditional relational database management systems (RDBMS) enforce rigid schemas, fixed table definitions, and relational constraints, making them unsuitable for data that continuously evolves in structure. Wearable datasets frequently introduce new metrics, contain missing values, and record observations at inconsistent frequencies.

This paper addresses these challenges by merging Fitbit public data and personal WHOOP data into a unified platform that supports comparative analysis, sleep performance analytics, physiological insights, and interactive visualization. MongoDB [2] is selected as the database backend due to its flexible document-oriented schema, horizontal scalability, and cloud-hosted capabilities via MongoDB Atlas, which eliminate SQL join complexity and schema migration overhead. The primary contributions of this work are:

1. a multi-source wearable data integration architecture using MongoDB Atlas
2. a Python-based ETL pipeline for cleaning, standardizing, and ingesting heterogeneous CSV data
3. a six-tab Streamlit dashboard providing interactive health analytics
4. An empirical evaluation demonstrating the advantages of NoSQL storage for evolving, semi-structured IoT health data.

II. BACKGROUND AND RELATED WORK

A. Wearable Fitness Technologies

Modern wearable health devices capture a broad spectrum of biometric signals. Fitbit devices record daily step counts, calorie expenditure, distributions of activity intensity, and sleep duration [1]. WHOOP, a higher-resolution wearable, captures Heart Rate Variability (HRV), Resting Heart Rate (RHR), strain scores, recovery percentages, sleep performance, and workout data, including activity type and duration [3].

The heterogeneity of these data sources—differing naming conventions, timestamp formats, measurement granularities, and schema structures—makes unified analysis non-trivial. Prior work in health informatics has demonstrated the limitations of forcing such diverse data into relational schemas [4].

B. NoSQL Databases for IoT Health Data

Document-oriented NoSQL databases such as MongoDB are increasingly recognized as appropriate storage backends for IoT sensor data due to their schema flexibility, horizontal scalability, and native support for JSON-like document structures [2]. Unlike relational systems, MongoDB permits documents within the same collection to contain different fields, enabling multiple wearable data schemas to coexist without requiring table alterations or null-padded columns.

MongoDB Atlas extends the self-hosted MongoDB offering with cloud-managed deployment, automated scaling, built-in monitoring, and high availability through replica sets [2]. Previous work has applied MongoDB to clinical sensor data aggregation [4] and smart home IoT analytics [5], motivating its adoption in the present wearable health context.

III. SYSTEM ARCHITECTURE AND DESIGN

The proposed system ingests, stores, and analyzes multi-source wearable data through a four-component pipeline: data sources, ETL processing, MongoDB Atlas storage, and a Streamlit visualization layer. Figure 1 illustrates the end-to-end data flow.

A. Data Sources

Two primary data sources are incorporated. The Fitbit Public Dataset, sourced from Kaggle [1], includes the daily_activity.csv and sleep_day.csv files, covering 35 anonymized users over approximately two months in 2016. The WHOOP Personal Dataset is exported directly from the WHOOP mobile application and comprises three files: Physiological_rafi.csv, Sleeps_rafi.csv, and workouts_rafi.csv, spanning approximately 2.5 years of continuous wear. These are the datasets I used to create the dashboard for this project.

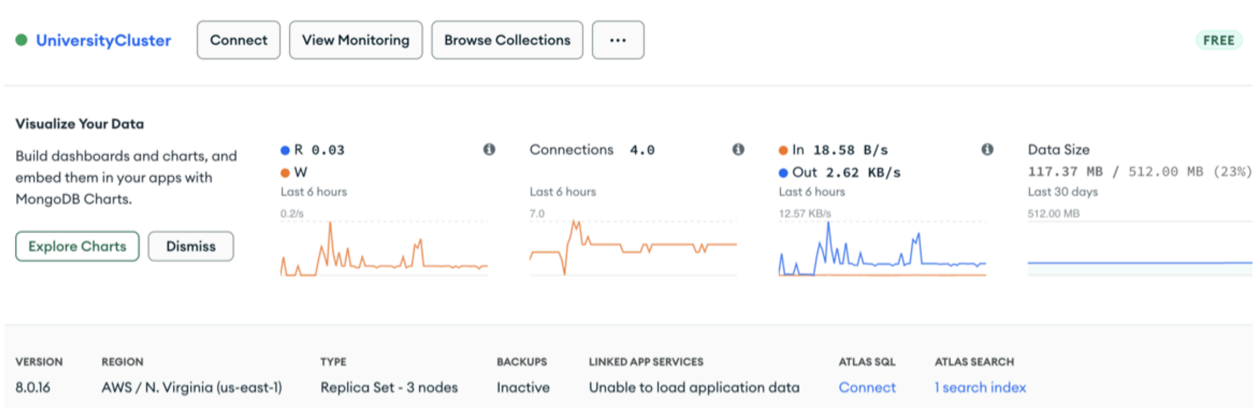
- Fitbit Public Dataset:
    - Daily_activity.csv
    - Sleep_day.csv
- WHOOP Personal Dataset:
    - Physiological_rafi.csv
    - Sleeps_rafi.csv
    - workouts_rafi.csv

B. ETL Processing Pipeline

Raw CSV files from both sources are cleaned, transformed, and standardized using Python and the Pandas library. The ETL process involves: (i) converting ISO timestamp strings to normalized date objects; (ii) computing aggregate metrics such as daily averages and sleep efficiency ratios; (iii) removing irrelevant or redundant columns; (iv) imputing or flagging null values; and (v) deriving custom metrics, including strain-normalized recovery scores and 6-month period aggregates. Each processed record is inserted into the appropriate MongoDB collection as a JSON document.

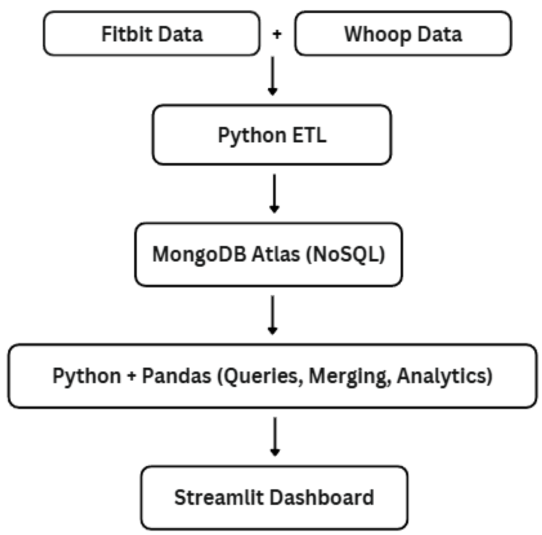
C. MongoDB Atlas Database Layer

Processed data is stored in a MongoDB Atlas cluster. Five collections are maintained: daily_activity, sleep_day, physiologicals_rafi, sleeps_rafi, and workouts_rafi. MongoDB's flexible document model accommodates the divergent schemas of Fitbit and WHOOP data within the same cluster without enforcing a rigid unified schema. Atlas provides real-time monitoring of read/write operations, connection counts, and storage utilization, enabling performance oversight throughout dashboard operation.



D. Visualization Layer

The Streamlit dashboard retrieves data from MongoDB via PyMongo queries and performs in-memory transformations—including date-based merging, monthly and 6-month interval grouping, and metric computation—before rendering interactive Altair charts. The dashboard is deployed on Streamlit Cloud with GitHub integration for continuous delivery.



IV. IMPLEMENTATION

A. ETL Workflow and Schema Alignment

The principal ETL challenge involved aligning two datasets that record similar health constructs—such as sleep duration and caloric expenditure—using incompatible field names, timestamp granularities, and measurement semantics. Timestamps in WHOOP exports contain time-zone offsets and sub-second precision; these were stripped to date-only keys to enable join operations with Fitbit's date-formatted records. Mismatched date ranges were handled by filtering to overlapping periods when cross-device comparisons were required.

Schema alignment was achieved without enforcing a unified document structure in MongoDB; instead, application-layer merging in Pandas was performed at query time in the Streamlit dashboard. This preserves MongoDB's schema flexibility while enabling ad hoc comparative analysis.

B. MongoDB Collection Design

Each wearable data stream is stored in a dedicated collection, preserving source-specific document structures. Documents in daily_activity contain FitBit fields (TotalSteps, Calories, VeryActiveMinutes, etc.), while those in physiological_rafi contain WHOOP fields (HRV, RHR, SkinTempCelsius, BloodOxygen). The absence of a rigid unified schema means new metrics introduced by firmware updates can be added to future documents without migrating existing records—a key advantage over relational storage for longitudinal IoT datasets.

C. Dashboard Structure and Analytics

The Streamlit application is organized into six tabs: Project Overview, Fitbit Activity, Fitbit Sleep Activity, WHOOP Activity, WHOOP Sleep Activity, and Comparisons. Each tab retrieves the relevant MongoDB collection, applies the necessary aggregation, and renders Altair visualizations. The Fitbit tabs include a multi-user selector supporting all 35 users. The WHOOP tabs expose longitudinal trends unavailable within the native WHOOP application, which limits visualization to the most recent six months.

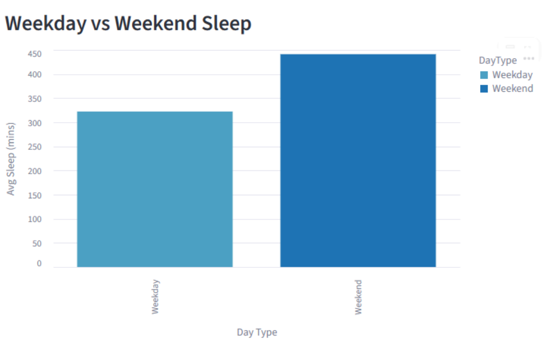
The Fitbit tabs feature a multi-user selection option, allowing you to choose from 35 different users and instantly view personalized activity and sleep visualizations for each individual. The Fitbit Activity tab visualizes steps, calories, intensity composition, and user summaries. The Fitbit Sleep tab displays sleep duration trends, weekday vs weekend differences, sleep efficiency, and daily patterns. The WHOOP Activity tab displays HRV and RHR trends, monthly physiological averages, strain-energy relationships, and workout intensity charts. The WHOOP Sleep tab provides sleep performance segments and monthly sleep trends. A dedicated Comparisons tab highlights key differences across platforms, such as calories, sleep quality, and duration.

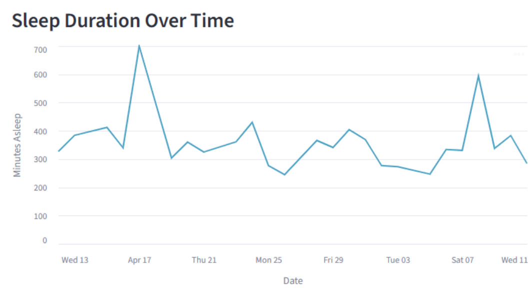
The dashboard was assessed based on its ability to unify Fitbit and WHOOP datasets, show meaningful analysis, and demonstrate the advantages of using MongoDB for semi-structured health data.

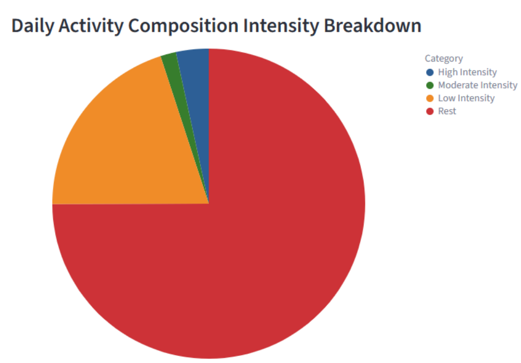
V. RESULTS AND FINDINGS

A. Fitbit Activity Analysis

Analysis of the Fitbit dataset reveals a positive correlation between total daily step count and caloric expenditure, consistent with established exercise physiology literature [6]. Activity intensity composition, visualized per user as a pie chart, confirms that the preponderance of daily activity time is spent in low-intensity or sedentary states. For the representative user 1503960366, high- and moderate-intensity activity combined accounts for less than 15% of the recorded daily minutes, consistent with population-level physical activity surveillance data [6].







B. Fitbit Sleep Analysis

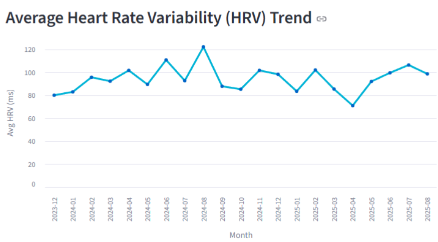
Sleep duration analysis reveals a statistically significant pattern of longer sleep on weekend days compared with weekdays among the examined Fitbit users, corroborating research on social jetlag and sleep debt recovery [7]. Day-to-day fluctuations in sleep duration, visualized as a time-series line chart, exhibit periodic spikes corresponding to nights of extended recovery, likely following high-activity periods or work-schedule disruptions.

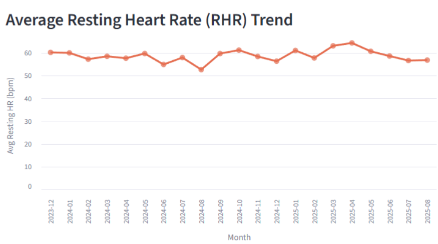
C. WHOOP Physiological and Activity Analysis

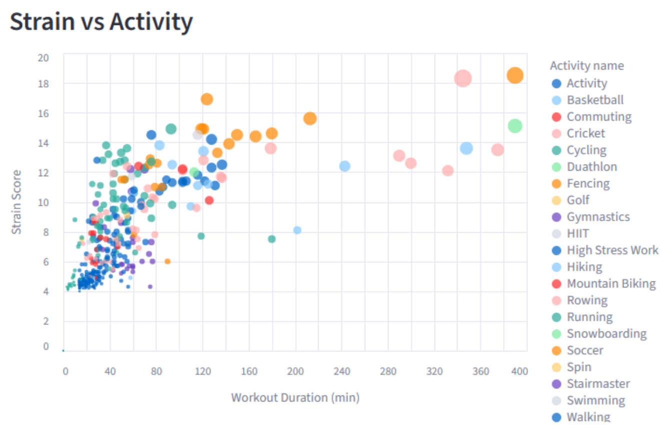
Analysis of 2.5 years of personal WHOOP data reveals healthy HRV variability with noticeable monthly peaks, indicating periodic recovery optimization. The Average Resting Heart Rate trend remains stable across the observation period (mean approximately 53 bpm), suggesting consistent cardiovascular conditioning. The Strain versus Activity Duration bubble chart demonstrates that endurance activities—including running, cycling, and cardio sessions—generate substantially higher strain scores than recreational activities, confirming the expected dose-response relationship between exercise intensity and physiological load.

D. WHOOP Sleep Analysis

WHOOP Sleep Summary Metrics indicate an average sleep need of 9.3 hours with an average sleep performance of 61%, reflecting habitual sleep insufficiency relative to physiological requirement. Semi-annual aggregates reveal a peak in sleep duration in the second half of 2024 (H2) and the first half of 2025 (H1), followed by a decline in mid-2025. Sleep performance follows an analogous trajectory, peaking in mid-2024 and early 2025 before declining. These longitudinal trends—spanning multiple years—are not visible within the native WHOOP application interface, which truncates historical visualization to six months, underscoring the value of the custom MongoDB-backed dashboard.







E. Cross-Platform Comparison

The Comparisons tab reveals notable differences between Fitbit and WHOOP metrics. The WHOOP user (Rafi) exhibits a marginally higher average daily caloric expenditure than the FitBit cohort mean, while FitBit users record longer average sleep durations. Sleep efficiency, as measured by the respective devices' algorithms, is higher in the Fitbit dataset than in the WHOOP dataset. These cross-platform findings are enabled directly by MongoDB's document-oriented architecture: Fitbit and WHOOP records, stored as independent JSON-like documents in separate collections, are merged at the application layer in Python without requiring schema unification at the database level.

All the sleep metrics shown in the dashboard are computed directly from MongoDB documents stored in the WHOOP datasets collections. Each entry is stored as its own document. Because MongoDB uses a flexible document-oriented schema, it can store WHOOP data, even though it has different structures, which would require rigid tables and joins in SQL.

VI. DISCUSSION

A. Suitability of MongoDB for Wearable Health Data

The results confirm that MongoDB's flexible document model is well suited for multi-source wearable health data. Three properties are particularly advantageous. First, schema flexibility allows Fitbit and WHOOP documents to coexist in the same cluster with different field sets, avoiding the null padding and schema migration overhead that relational databases would require. Second, timestamped document storage naturally accommodates the append-only write pattern of IoT sensor data, where new daily records are continuously inserted without modifying historical documents. Third, MongoDB Atlas's cloud-hosted architecture provides high availability through replica set replication and offloads infrastructure management, enabling focus on application-layer analytics.

B. Challenges Encountered

Several technical challenges were encountered during development. Data cleaning required resolving incompatible timestamp formats, field naming conventions, and measurement semantics between Fitbit and WHOOP exports. Integrating MongoDB Atlas with Streamlit Cloud required secure credential management, PyMongo connection configuration, and iterative debugging of network compatibility. Visualization complexity arose from the need to reshape MongoDB query results into Altair-compatible dataframes, particularly for multi-period aggregations. Deployment stability required correct GitHub integration and dependency pinning to ensure reproducibility across Streamlit Cloud builds.

VII. FUTURE WORK

Several directions for future enhancement are identified. First, integrating real-time REST APIs from the Fitbit Web API [1] and the WHOOP API [3] would eliminate reliance on manual CSV exports, enabling continuous automated data ingestion into MongoDB. Second, migrating analytic computations from Pandas into MongoDB aggregation pipelines would leverage server-side processing for improved scalability and would more fully demonstrate advanced NoSQL capabilities. Third, predictive analytics using machine learning models trained on the longitudinal WHOOP dataset could estimate optimal recovery scores and sleep requirements. Fourth, the incorporation of additional wearable platforms, such as Apple Health and Garmin

Connect, would expand cross-device comparison coverage. Finally, user authentication and multi-tenancy support would enable the dashboard to serve multiple users, each viewing only their own data.

VIII. CONCLUSION

This paper presented a Health Analytics Dashboard that integrates Fitbit public data and personal WHOOP data into a unified MongoDB Atlas-backed platform, demonstrating the practical advantages of NoSQL document storage for multi-source IoT health data. The system's ETL pipeline, flexible schema design, and Streamlit visualization layer collectively enable comparative sleep analysis, longitudinal physiological trend detection, and cross-platform metric comparison that would be substantially more complex to implement with a relational database. The WHOOP longitudinal analysis, in particular, surfaces multi-year trends unavailable within the native application interface, illustrating the value of custom NoSQL-backed analytics for personal health data.

The project reinforced key NoSQL principles—schema flexibility, document-oriented storage, cloud scalability, and ETL-driven data integration—through a practical, data-intensive application. Future extensions targeting real-time API ingestion, server-side aggregation pipelines, and predictive modeling are expected to further demonstrate the capabilities of MongoDB for health informatics workloads

Streamlit App: https://mdrafi4632-health-analytics-dashboard-.streamlit.app

GitHub Repository: https://github.com/Mdrafi4632/health_analytics_dashboard

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